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# The Art of Post Captions: Readability and User Engagement on Social Media

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## Abstract

In addition to the significance of visual content, text characteristics serve as salient factors influencing how users react to social media posts. By extracting 9,766 Instagram posts published by destination marketers, this research investigates the interplay between the readability of post captions and destination attributes derived from pictorial content on user engagement. Grounded in the theoretical lens of processing fluency and image-text congruity, the findings revealed that vibrant spots/activities with simpler texts lead to a significantly higher engagement rate, while cultural and historical attractions that use more complex texts are unaffected. Yet, when complex texts are applied to spots that are neither vibrant nor cultural, user engagement decreases significantly. Overall, this research contributes to the interdisciplinary discourse on linguistics and psychological esthetics in destination marketing relating to computer-mediated environments. Regarding practice, the findings provide insights into the effectiveness of different levels of readability across diverse tourism marketing settings.

## Keywords

readability, user engagement, visual analysis, processing fluency, data analytics

## Introduction

While commonly-adopted marketing strategies such as the use of pictures, short films, and videos have been utilized to enhance destination promotion (Ying et al., 2022), texts play an irreplaceable role when it comes to social media (Gkikas et al., 2022). They facilitate communication exchange and provide informative content (Hooker & Cooper, 2022), adding a critical dimension to overall marketing efforts. By integrating the discipline of linguistics with the concept of destination marketing in social media, the synergy of being informative, attractive, and readable has been found to render content compelling (Patel et al., 2020). In fact, the strategic promotion of destinations relies heavily on readability. Readability, in this context, refers more so to forging connections with viewers in an accessible manner than to the outcome of good writing skills or the sophistication of an argument (Fang et al., 2016).

Since communication presupposes comprehension, the need for educators in the realm of English teaching, whether as a first language or a second/foreign language, has accelerated the development of readability measurements (Greenfield, 2004). Nowadays, readability in digital marketing encompasses hooks and should be customer-focused (Gkikas et al., 2022). Previous studies reported that readability not only affects the level of understanding of promotional messages from travel agencies (Li et al., 2023) but also influences perceived value of attraction (Fang et al., 2016) and triggers engagement on social media (Gkikas et al., 2022). The level of education required is even associated with perceptions of a destination (Sattari & Wallström, 2013). Hence, to evaluate destination marketing on social media based on textual content, marketers have adopted various readability metrics, including the length of a text as well as its character count (Gkikas et al., 2022), the Dale-Chall formula, Flesch Kincaid reading ease, and textual lexical diversity (Aakash & Gupta Aggarwal, 2022; McCarthy & Jarvis, 2010), amongst others.

In destination marketing, the theory of processing fluency plays a significant role in understanding how tourists perceive and engage with content ( L. R.Tang et al., 2014). Originating from the field of psychological esthetics, this theory explains how the subjective ease or fluency of

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information affects human cognitive processing (Reber et al., 2004). For instance, the processing fluency of travel websites is associated with one's attitude toward destinations ( L. R.Tang et al., 2014), and similar results have also been found with respect to tourism marketing messages (Hanks et al., 2016). Consequently, offering easy-to-read content on social media is more likely to lead to positive reactions and user engagement (Davis et al., 2019).

However, despite the significance of readability within the arena of social media (Pancer et al., 2019), pictorial elements have received more attention in destination marketing (Philp et al., 2022; Yu & Egger, 2021) than text-related characteristics (Gkikas et al., 2022). In addition, notwithstanding that recent scholars have initiated a call to examine processing fluency in digital environments (Davis et al., 2019; Gkikas et al., 2022; Hong et al., 2023; Pancer et al., 2019), the focus continues to remain on longer text-based content such as online reviews and website information (Aakash & Gupta Aggarwal, 2022; Filieri & McLeay, 2014; Li et al., 2023). The fact that the readability of social media texts has been largely ignored (Davis et al., 2019) is problematic as tourists increasingly rely on these short and unstructured pieces of content to make traveling decisions (Ismarizal & Kusumah, 2023). Furthermore, there is a lack of empirical evidence on the impact of processing fluency, particularly concerning short captions (Pancer et al., 2019) that hold great relevance in destination marketing (Conti & Cassel, 2021). These short captions are significant due to their prevalence on social media platforms like Instagram, which is the most-downloaded app globally (Newberry, 2023) and a major channel adopted by destination marketers (Hooker & Cooper, 2022; Ismarizal & Kusumah, 2023).

Intrigued by the power of readability to boost users' engagement (Davis et al., 2019), this research aims to investigate the interplay between reading ease and the complexity of texts based on different types of destination attributes derived from social media pictures as well as their impact on user engagement. In general, this research bridges linguistics and destination marketing by advancing the understanding of readability (Newbold & Gillam, 2010) and the literature on processing fluency (Reber et al., 2004) in the social media landscape. By means of data analytics, the findings provide guidance for designing readable captions that align with destination attributes. Such guidance is not only beneficial for marketers but for other stakeholders involved in promoting tourism on social media platforms.

## Literature Review

### *Destination Marketing on Instagram*

Instagram is a visual-based social media platform that is especially popular among younger generations, with 61% of its audience falling into the age range of 18 to 34 (Newberry, 2023). It is considered an effective platform to market tourist destinations due to the easiness of sharing photographs and information (e.g., geographical location) as well as the applicability of other functions to encourage user interactions (Filieri et al., 2021). Owing to its visual-based characteristics, a substantial body of literature has focused on the impact of visual elements on destination marketing (e.g., Philp et al., 2022), such as how the colors of pictures attract user engagement (Yu & Egger, 2021). Comparatively, studies on caption analyses (e.g., Conti & Cassel, 2021; Hooker & Cooper, 2022) are lacking. By directly demonstrating destination attributes, visual content can boost tourists' cognitive (i.e., knowledge and beliefs), affective (i.e., emotions), and conative (i.e., intentions) aspects (Eagly & Chaiken, 1993), thus increasing travel intention. Meanwhile, captions allow readers to make sense of the visual content (Hooker & Cooper, 2022) to make informed decisions regarding their travels. While the presence of visual aids embellishes the marketed destination (Yim et al., 2021), Instagram captions can greatly influence tourists' perception toward a destination (Conti & Cassel, 2021) and are ultimately important determinants of one's destination choices (Hooker & Cooper, 2022; Ismarizal & Kusumah, 2023).

In fact, the uniqueness of social media content for destination marketing lies in the synergy of combining both images and texts (Egger & Yu, 2022). Since image-centric platforms such as Instagram thrive on visual content, short captions serve as an ideal complement to the impact of graphics (Conti & Cassel, 2021). Thus, by blending visible allure with highly readable texts, they encourage the quick and easy consumption of content, allowing travelers to promptly grasp the essence of a destination (Hooker & Cooper, 2022). Scholars further reinforced that tourists are more likely to choose a destination after obtaining clear and straightforward information ( L. R.Tang & Jang, 2011), potentially because they alter the perceived attractiveness and quality of said destination (Qi et al., 2008).

In this context, the official accounts of tourism organizations play a particularly crucial role, especially since Generation Y and Z tourists rely heavily on the captions of official accounts when making travel decisions (Ismarizal & Kusumah, 2023). At the same time, among the various forms of post captions, short narrative captions used by official accounts are exceptionally powerful in increasing tourists' visit intentions to the marketed destination (Hooker & Cooper, 2022). Nevertheless, research on the impact of Instagram captions, and the underlying reasons of said impact, remains insufficient. Taking into consideration average users' short attention span (7–15 seconds) and reading fluidity, it has been recommended that Instagram captions contain a narrative of no longer than 125 characters (J. S.Zhang & Su, 2023). Short stories are the easiest to process, and high readability results in increased user engagement (Gkikas et al., 2022), which researchers explain in terms of processing fluency (Reber et al., 2004).

## The Role of Processing Fluency in Linguistics and Tourism

Stemming from the fields of cognitive neuroscience and psychology, processing fluency is a theory referring to the ease with which individuals process and understand information (Reber et al., 2004). It has been the subject of abundant research in psychology (Reber et al., 2004), linguistics (Nunes et al., 2015), and destination marketing ( L. R.Tang & Jang, 2011). One of the earliest studies on processing fluency uncovered that individuals are more likely to remember information that is presented in an easy-to-read format, compared to information that is rather difficult to comprehend (Jacoby & Dallas, 1981). This effect can be attributed to one's cognitive bias, referring to the tendency of people to judge the truth of a statement based on how easily they can comprehend the information ( K. Y.Wang et al., 2013). Thus, when the brain is presented with jargon-laden language, one's decision-making process, perceptions, and experiences can be unconsciously affected (Brouillet et al., 2023; Hong et al., 2023).

The role of processing fluency has also been examined in prior studies on reading comprehension, demonstrating that readers are more likely to comprehend and remember information that is presented in a well-organized format ( T. J.van Rompay et al., 2009). Since text comprehension is highly interactive (de Vries et al., 2012), reading materials presented in a fluent manner should lead to a deeper level of engagement (Davis et al., 2019), for example, through generating inferences and making connections between ideas. Even when evaluating language, fluency interconnects with perceived accuracy and trustworthiness (Schwarz et al., 2021); therefore, to influence processing fluency, content creators may also utilize familiar and unfamiliar words or phrases in social media posts (Pancer et al., 2019). Materials structured on more frequently occurring words may enhance one's perceived trust toward the content and encourage the user's willingness to process the information (Schwarz et al., 2021). Certainly, however, metacognitive experiences can vary depending on the consumption domain (Schwarz et al., 2021). Knowledge from marketing suggests that unfamiliarity, serving as a proxy of uniqueness, can signal greater value in a special occasion (vs. an everyday life setting) (Pocheptsova et al., 2010).

In destination marketing, processing fluency has been found to have an effect on a variety of language-related behaviors, including the reading comprehension of travel websites (L. R.Tang & Jang, 2011), tourists' responses to hotel messages (Hanks et al., 2016), and the evaluation of materials (Reber et al., 2004). Simultaneously, other scholars have stated that processing fluency may affect the image of a destination (L. R.Tang et al., 2014). Hence, by understanding the role of processing fluency in tourism, marketers can increase the appeal and perceived quality of their offerings while ultimately improving satisfaction. With the digitalization of marketing environments nowadays, one way to leverage processing fluency is to assess the readability of presented information via various measurements (Gkikas et al., 2022; Pancer et al., 2019).

## Readability Features and Measurements

The characteristics of text readability encompass numerous linguistic levels (Gkikas et al., 2022) including vocabulary choice, word length, sentence length, structural complexity, and lexical richness. To determine the effectiveness of written communication in influencing one's ability to engage with presented information (Davis et al., 2019), the Dale-Chall readability formula (Dale & Chall, 1948) is viewed as perfectly designed for short content on social media (Pancer et al., 2019). It measures word familiarity and syntactic complexity by referring to a dictionary-based list of terms (Narman et al., 2018). By turning the score into U.S. grade levels, representing the different stages of students' academic journeys spanning across 12 grades, a study analyzing the readability of Facebook posts revealed that the Dale-Chall score for simple posts was 2.0 (Grade 2 level), whereas the score for posts containing more difficult words was 5.0 (Grade 5 level) (Pancer et al., 2019).

Lexical diversity, which refers to the number and variety of words used in a text (McCarthy & Jarvis, 2010), serves as another salient factor determining the effectiveness of textual content (Koizumi & In'nami, 2012). Assessments include measure of textual lexical diversity (MTLD) and hypergeometric Distribution D (HDD) (McCarthy & Jarvis, 2010); yet, MTLD has been found to be more suitable to analyze short-text data (Koizumi & In'nami, 2012).

To date, researchers have applied readability metrics to evaluate Facebook posts (Gkikas et al., 2022), tourism websites (Sattari & Wallström, 2013), and online community exchanges on Stack Overflow (Al Qundus et al., 2020). Nevertheless, although state-of-the-art knowledge indicates the role of lexical diversity in influencing readability (Jian et al., 2022; Koizumi & In'nami, 2012), little findings exist when it comes to the readability of short messages (Gkikas et al., 2022) highly relevant to destination marketing. Across the broader marketing spectrum, prior research has shown that consumers perceive products or services that are advertised using more sophisticated language as being of higher quality (Morales et al., 2012). Some marketers traditionally attributed this to the idea of image congruence; that is, if the product, the marketing method, and the consumers' self-concept align to their level of sophistication, higher purchase and recommendation intentions emerge (Belanche et al., 2021; T.Zhang et al., 2019). However, the use of complex language can also make marketing messages more difficult to understand, leading to decreased comprehension (Davis et al., 2019; Hong et al., 2023).

To strike a balance between the introduction of sophisticated language and the need for clarity, marketers are encouraged to strategically increase lexical diversity while maintaining readability and simplicity (Garett et al., 2016). Such a balance is especially important in tourism (Li et al., 2023)—a field consisting of diverse target consumers. As an example of strategic simplification, synonyms based on the most frequently used English words could be used to add variety to written information. Suggested by educators, leveraging the most frequent words in both spoken and written English can help ensure that the majority of an audience understands the message (Narman et al., 2018).

### Readability and User Engagement on Social Media

Although texts go hand in hand with visual elements in destination marketing, current mainstream research mostly focuses on the interplay between photographs and tourists' reactions on social media (Yim et al., 2021; Yu & Egger, 2021). Yet, from a neuroscience perspective, it is the features of natural language that affect text readability, which subsequently influence how readers engage with information (Jian et al., 2022). Knowing that social media posts are often short and unstructured (Egger & Yu, 2022), scholars have initiated a call to assess how readability impacts the effectiveness of social media communication (Al Qundus et al., 2020; Gkikas et al., 2022; Pancer et al., 2019). Thus far, existing literature has demonstrated that easy-to-read messages often enhance engagement (e.g., liking behavior) (Gkikas et al., 2022), positive attitude (Hanks et al., 2016), and information sharing (Aakash & Gupta Aggarwal, 2022).

In addition to writing style, the complexity of the language used is equally important (Davis et al., 2019), for example, the presentation of passive voice. As a significant predictor of psychological functioning (e.g., psychological distance) (Sepehri et al., 2022), passive structures may conceal the identity/personality of the sender (Smeuninx et al., 2020), thereby resulting in a decrease in social media engagement. In destination marketing, relevant research discovered that the degree of readability implicitly links to tourists' perceptions of a destination and can even influence one's imagination of the experience (Sattari & Wallström, 2013). Such phenomena can be traced back to human metacognition in psychology, highlighting the association between the ease of processing and the positive impact it has on social media (Pancer et al., 2019).

To transcend the status quo of knowledge mainly generated from different types of posts and a text's level of vividness (Sreejesh et al., 2020), readability, which functions as a compelling antecedent of user engagement, merits further attention (Pancer et al., 2019). Notably, depending on the nature of social media platforms, the effect of readability can differ; for instance, while there is a positive relationship between readability and consumer reactions on Facebook (Pancer et al., 2019), a study conducted by Davis et al. (2019) revealed the opposite effect on Twitter, potentially because consumers expect additional information shared by businesses. Though knowledge derived from the above text-centered platforms can serve as a benchmark, they become a major issue in destination marketing, which relies heavily on the integration of visual elements and short texts (e.g., on Instagram). Thus, an urgent need arises to explore destination marketers' textual information characteristics to design engaging and effective materials for potential tourists.

## Methodology

To investigate the interaction between readability metrics and destination attributes derived from pictorial content on user engagement toward social media posts published by destination marketers, the following steps were taken in this study.

### Step 1: Data Extraction

Considering the relationship between tourism marketing campaigns of destination marketing organizations (DMOs) and the overall economic impact in promoting and attracting tourists, this research focuses on the Group of 20 member nations, where tourism revenue plays a significant role in the national gross domestic product (GDP), to examine the correlation between destination marketing campaigns by DMOs and their economic impact (Pratt et al., 2010). According to the economic impact report released by the World Travel & Tourism Council (2022), USA, China, Germany, Japan, and Italy had the highest tourism contribution to GDP and were thus selected as the research context. The selection was cross-checked against the cognitive image attributes outlined by Taecharungroj and Mathayomchan (2021) to ensure a diverse representation of broad attribute types. For example, the USA is linked with mountain and river, China with people and events, Germany with city and building, Japan with street and plants, and Italy with architecture and sea, among other associated characteristics (Taecharungroj & Mathayomchan, 2021).

Next, marketing activities performed on Instagram, which functions as one of the leading social media platforms for promoting destinations (Filieri et al., 2021), were evaluated by extracting all available posts published on the official Instagram accounts of the respective marketers—managed by the national tourist board (i.e., @visittheusa, @visitchina, @germanytourism, @visitjapanjp, and @italiait). Data extraction was conducted in December 2022, encompassing posts dating back to 2013. On average, there were 1,233 posts per year spanning from 2013 to 2022. Extracted data included post captions, number of likes and comments, number of account followers, and post and image URLs, leading to a total of 13,539 items. To meet the criteria of the subsequent analysis (explained below), however, video-based posts were excluded and only the first picture was taken from carousel posts containing more than one photo, resulting in 12,797 Instagram posts overall.

### *Step 2: Image Annotation and Clustering*

Since marketing information for different tourism activities and experiences can differ widely, post captions vary in terms of their level of complexity. Hence, a key step in the investigation is to cluster the extracted posts into distinct categories with similar offerings based on their photographic content. This approach recognizes pictures as the initial hook for users and emphasizes the need to consider their impact alongside the accompanying textual content. By prioritizing pictorial elements, it facilitates the understanding of how caption readability influences engagement.

Specifically, pictures were downloaded based on the previous extracted image URLs, and image annotation was performed using Google Cloud Vision API (Philp et al., 2022). For each picture, those entities from the top image labels with $>0.5$ confidence level were presented (Chen & Chen, 2017). These labels were subsequently transformed into numbers using document embeddings (Egger, 2022), which means that words appearing closer in a vector space connotate similar meanings. The procedure continued by applying the Louvain algorithm, a heuristic algorithm for community detection in large networks, to cluster images according to their detected labels. The algorithm operates by iteratively improving the modularity of the network division to find clusters of densely connected nodes within the network (Nguyen et al., 2018). In addition to its application in the identification of communities in social networks, the Louvain algorithm has emergingly been adopted as a technique for image classification (Nguyen et al., 2018) in tourism marketing (Yu & Egger, 2021).

### *Step 3: Data Pre-processing and Preparation*

Different from conventional text analytics that often involve the removal of stop words, in this research, the post captions remained mainly unaltered owing to the particular focus on readability. Thus, in the pre-processing step, only hashtags included at the end of a post, references to users (e.g., @ username), and emojis were removed. Emojis are a highly complex issue in themselves as they are categorized differently by diverse researchers (e.g., X.Wang et al., 2023), and their impacts on user engagement are far from unanimous (Manganari et al., 2020). Hence, to avoid complicating the current investigation, emojis were not considered further.

To ensure the completeness of sentences, hashtags used as part of a word or phrase within sentences were kept (e.g., This #hashtag was included.), whereas non-English posts were excluded. The final dataset therefore consisted of 10,852 posts in total. As a next step, the engagement rate for each post was calculated from the extracted data related to the number of likes and comments. Being the most common metric for evaluating marketing performance on Instagram, user engagement is computed by adding the likes and comments of a post and dividing it by the given number of followers of an account (Stevanovic, 2022).

### *Step 4: Readability Measurement*

Following this, the study measured readability on two levels: (1) structural complexity and (2) lexical complexity. The former focuses on sentence-level complexity of the grammatical structure of a language, while the latter refers to the complexity of the vocabulary used in texts (Saricaoglu & Atak, 2022). For this research, several metrics were selected to enhance the robustness and validity of the results. Measurements for structural complexity included the Dale-Chall readability formula as well as passive voice construction, and lexical complexity was assessed via MTLD and HDD for lexical diversity (McCarthy & Jarvis, 2010). The results of MTLD and HDD were averaged for better measurement accuracy ( Y.Liu et al., 2021). All analyses were performed using Python.

Inspired by the Flesch-Kincaid readability test, the Dale-Chall algorithm utilizes the percentage of difficult words and sentence length to calculate the readability of a text (Pancer et al., 2019). By determining the difficulty of words using a list of 3,000 terms—which at least 80% of fourth-grade American students are familiar with—it transforms the scores to grade levels. While the metric does not have a fixed range, it typically varies from four or lower to 16 or higher. A higher index indicates that a higher level of education is required. For instance, a value of 9 represents 13th grade. Furthermore, since readability also relates to linguistic style, and applying passive constructions tends to decrease user engagement, the passive voice was analyzed based on a post caption's proportion of passive sentences to total number of sentences (Sepehri et al., 2022). Concerning lexical complexity, MTLD reflects the diversity of vocabulary used in a text, which is calculated by dividing the total number of unique words by the total number of words in a text (McCarthy & Jarvis, 2010). A high lexical diversity implies that a text contains a broader and more varied range of vocabulary terms. Similarly, HDD is designed to assess the richness of one's vocabulary based on a combination of quantitative and qualitative measures, for instance, measures of word frequency, word length, and word class (McCarthy & Jarvis, 2010). The underlying reason thereof is the fact that the size and diversity of an individual's vocabulary play a major role in their ability to communicate effectively. The results of MTLD and HDD were averaged for better measurement accuracy ( Y.Liu et al., 2021).

**Table 1.** Summary of Image Clusters and Readability Metrics.

| Image cluster | *n* | Structural | | Lexical | |
|---|---|---|---|---|---|
| | | DC | PS | LD | EN |
| Culture, history, and architecture | | | | | |
| 1: Artworks and indoor environment | 1,243 | 12.1 (2.59) | −0.13 (0.30) | 67.4 (44.1) | −2.64 (0.88) |
| 2: Tower, plaza, and Gothic design | 891 | 12.5 (2.69) | −0.18 (0.35) | 75.1 (42.4) | −2.35 (0.63) |
| 3: Building landscape | 548 | 11.7 (2.59) | −0.16 (0.32) | 65.4 (41.8) | −2.31 (0.63) |
| 4: Alleyway and street in the city | 473 | 13.1 (2.64) | −0.17 (0.35) | 83.4 (41.5) | −2.22 (0.51) |
| 5: Sacred places | 328 | 12.1 (2.66) | −0.16 (0.34) | 75.6 (43.7) | −2.40 (0.70) |
| 6: Monumental architecture | 259 | 12.9 (2.70) | −0.13 (0.32) | 82.5 (46.9) | −2.34 (0.65) |
| Vibrant spots and activities | | | | | |
| 7: Grassland, forest, and wildlife with people | 556 | 11.0 (2.19) | −0.15 (0.30) | 61.3 (37.0) | −2.36 (0.66) |
| 8: Means of transport | 309 | 11.2 (2.35) | −0.07 (0.21) | 54.7 (38.1) | −2.72 (0.87) |
| 9: Food and beverage | 296 | 10.7 (1.89) | −0.10 (0.26) | 52.0 (37.6) | −3.11 (0.87) |
| 10: Water activities | 128 | 11.8 (2.73) | −0.11 (0.28) | 67.4 (42.6) | −2.49 (0.70) |
| 11: General travel photography with people | 118 | 11.4 (2.25) | −0.08 (0.25) | 53.3 (39.3) | −2.86 (1.01) |
| Natural scenery | | | | | |
| 12: Greenery with buildings and objects | 1,072 | 11.6 (2.44) | −0.16 (0.31) | 66.1 (39.5) | −2.33 (0.64) |
| 13: Atmospheric moods | 852 | 11.9 (2.52) | −0.16 (0.32) | 70.2 (40.8) | −2.30 (0.63) |
| 14: Coastal and water impressions with cityscape | 828 | 12.2 (2.58) | −0.15 (0.32) | 72.1 (39.9) | −2.30 (0.59) |
| 15: Seascape and nature | 722 | 12.7 (2.63) | −0.16 (0.33) | 79.7 (40.8) | −2.35 (0.59) |
| 16: Mountain landscape | 575 | 12.0 (2.49) | −0.15 (0.32) | 68.6 (39.4) | −2.43 (0.66) |
| 17: Flora and fauna | 568 | 11.7 (2.22) | −0.18 (0.32) | 68.8 (38.0) | −2.27 (0.65) |

*Note*. DC = Dale-Chall; PS = passive sentences % (log); LD = lexical diversity; EN = engagement rate (log).

### *Step 5: Analysis of Readability and Engagement*

To investigate the effect of readability on engagement rate, the analysis began with assumption checks for a multiple regression analysis. First, descriptive inspection showed that readability measures contained outliers. The application of an isolation forest followed in order to detect said outliers with a contamination of 10% for all variables (F. T.Liu et al., 2008). After the removal of outliers, the final dataset contained 9,766 posts. Next, since an inspection of the Q-Q plots showed that engagement rate and passive percentages failed to meet the requirements, a log(10) was performed on these variables. The plots were deemed acceptable after transformation of these variables, and all requirements for multiple regression were met. Finally, to control for country effects, dummy variables were introduced for each country, together with the dependent variables.

In light of the identified image clusters, first, to indicate that different levels of complexity are applied to language across countries and various tourism offerings, ANOVA was performed to report country and cluster differences against the three readability metrics (i.e., Dale-Chall, passive sentence percentage, and lexical diversity). Thereafter, correlation analysis was conducted in general between the measures, followed by regression analysis between readability metrics and user engagement to determine whether structural complexity or lexical complexity influence user engagement depending on the different tourism offerings on social media.

## Results

### *Image Clustering and Descriptive Statistics*

Starting with image classification, the Louvain algorithm was able to generate 17 image clusters. Table 1 presents an overview of the identified image topics with their corresponding readability scores and engagement rate. Subsequently, human visual inspection was conducted on the designated pictures (Shen et al., 2016), together with the image labels containing the highest term frequency-inverse document frequency, to facilitate the naming process of each cluster (Yu & Egger, 2021). Additionally, to allow for a systematic and constructive discussion, the results from the topic modeling algorithms were organized into three major groups (Table 1), considering that some topics may exhibit similarities with overlapping characteristics (Egger & Yu, 2022).

Based on the identified topics, the results suggested several cultural and historical spots such as "monumental architecture," "tower, plaza, and Gothic design," and "artworks and indoor environment" as well as some vibrant (i.e., lively and exciting) spots and activities such as "water activities," "grassland, forest, and wildlife with people," and "food and beverage." The findings also revealed pictures related to natural scenery including "seascape and nature," "mountain landscape," and "flora and fauna." Overall, it appears that "alleyway and street in the city," "flora and fauna," and "coastal and water impressions with cityscape" achieved the highest engagement rate, whereas users engaged less with topics involving "means of transport," "general travel photography with people," and "food and beverage." Yet, while the informativeness of post captions or the esthetics of photographs may influence user engagement, this research focuses on the synergy between text readability and destination attributes derived from pictorial content.

### *Main Analysis*

A Pearson's correlation was run to assess the relationship between the Dale-Chall score, lexical diversity, and the use of the passive voice. The findings implied a significant and positive relationship between the Dale-Chall score and lexical diversity ($r = .535$, $p < .001$) and between lexical diversity and the use of passive sentences ($r = -.057$, $p < .001$). Nevertheless, the correlation size between the latter items is negligible. A correlation between the Dale-Chall score and lexical diversity, on the other hand, was expected as the Dale-Chall formula adopts both sentence length and word difficulty (Dale & Chall, 1948). Such results are consistent with earlier research, stating that word difficulty, also known as lexical sophistication, and lexical diversity had high correlations (Išková, 2012). Subsequent regressions revealed that the Variance Inflation Factor between the Dale-Chall formula and lexical diversity was a mere 2.5—below the recommended threshold of 5 (Menard, 2001); hence, it can be claimed that the three language variables are distinct from one another.

The readability metrics across countries and destination marketing offerings were examined next, with a one-way ANOVA indicating a statistically significant interaction between country and image clusters for the Dale-Chall readability score ($F(1, 9681) = 2.43$, $p < .001$, partial $\eta^2 = 0.016$) and lexical diversity ($F(1, 9681) = 1.89$, $p < .001$, partial $\eta^2 = 0.012$). While a statistically significant interaction between country and image clusters for the use of passive sentences could not be determined ($F(1, 9681) = 1.02$, $p = 0.433$, partial $\eta^2 = 0.007$), the analysis of the main effect for country was found to be statistically significant ($F(4, 9681) = 50.41$, $p < .001$, partial $\eta^2 = 0.020$).

To uncover the relationship between structural complexity and user engagement, a multiple linear regression was calculated to predict engagement rate using the Dale-Chall scores and the passive voice. Principally, the results explained that both measurements ($p_{\text{Dale-Chall}} < 0.001$; $p_{\text{passive voice}} < 0.001$) served as significant predictors of engagement rate, regardless of the image clusters. Yet, when taking destination attributes from the results of the image clustering into account, the findings revealed that merely the Dale-Chall metrics significantly influenced user engagement in clusters 1 (artworks and indoor environment, estimate $= -0.05$, $p < .001$), 5 (sacred places, estimate $= -0.07$, $p < .01$), 8 (means of transport, estimate $= -0.09$, $p < .01$), and 11 (general travel photography with people, estimate $= -0.09$, $p < .05$), whereas the use of passive sentences was a significant predictor of engagement in clusters 3 (building landscape, estimate $= -0.19$, $p < .05$), 13 (atmospheric moods, estimate $= -0.14$, $p < .05$), and 17 (flora and fauna, estimate $= -0.15$, $p < .05$).

As for lexical complexity, similarly, without considering the feature of pictorial content, the results implied that lexical diversity was a significant predictor of engagement rate ($p < .001$). Nonetheless, this was only the case for clusters 1 (artworks and indoor environment, estimate $< 0.00$, $p < .001$), 4 (alleyway and street in the city, estimate $< 0.00$, $p < .05$), 8 (means of transport, estimate $< 0.00$, $p < .05$), 9 (food and beverage, estimate $< 0.00$, $p < .001$), and 11 (general travel photography with people, estimate $= 0.01$, $p < .01$). Table 2 provides an overview of the results of this study's regression analyses.

## Discussion

Tourism is undoubtedly an inherently visual field, and leveraging pictorial content is a quintessential element in destination marketing on social media (He et al., 2022; Yu & Egger, 2021). Given the diversity of tourism destinations, ensuring homogeneity across photos and graphics thus becomes essential as it focuses on specific image groups, facilitating a more coherent examination of the content. In line with existing research, the most frequently featured posts appear to be of natural impressions (e.g., greenery, mountains, seascape) (Picazo & Moreno-Gil, 2019), constituting the highest number of pictures shared by marketers within the current dataset. Other popular topics include artwork and indoor environments, followed by architectural-related images (e.g., towers, building landscapes), culture (e.g., sacred places), and gastronomy, amongst others. While acknowledging that images simplify complex issues through structuring designs ( T. J. L.Van Rompay et al., 2010), post captions play a crucial role alongside visuals in destination marketing (Conti & Cassel, 2021; Hooker & Cooper, 2022). These captions must be read and understood to provide users with more descriptive and informative content (de Vries et al., 2012). By using a variety of readability measures, the aim of this research is to investigate the synergistic effect of text readability and visual elements on user engagement.

Overall, this study revealed that marketers tend to combine simpler sentences and lexical items with more vibrant activities/spots (e.g., food and beverage, means of transport, water activities) than those that are viewed as cultural or historical (e.g., alleyway, monumental architecture, gothic designs). Such practices are in line with the congruity theory (Li et al., 2023; Osgood & Tannenbaum, 1955), implying that marketers might be aware of how different tourism destinations/activities target different populations and that the level of sophistication in the description should match the image of the product. However, from the perspective of potential tourists (i.e., Instagram users), generally, the current study's findings disclosed that complex sentence structures have a negative impact on user engagement but

**Table 2.** Estimation Results of Regression Analysis.

| | | All | C1 | C2 | C3 | C4 | C5 | C6 | C7 | C8 | C9 | C10 | C11 | C12 | C13 | C14 | C15 | C16 | C17 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Structural complexity | Adj. $R^2$ | .21 | .33 | .21 | .12 | .19 | .17 | .15 | .18 | .18 | .34 | .12 | .19 | .15 | .21 | .54 | .24 | .22 | .41 |
| | Intercept | −2.63*** | −2.84*** | −2.53*** | −3.01*** | −2.80*** | −2.60*** | −2.64*** | −2.95*** | −2.55*** | −2.79*** | −2.71*** | −2.23*** | −2.30*** | −2.16*** | −3.50*** | −3.42*** | −2.51*** | −2.63*** |
| | China | 0.83*** | 1.27*** | 0.67*** | 0.81*** | 0.54*** | 0.54*** | 0.46** | 0.96*** | 0.69*** | 0.71*** | 0.73** | 0.54*** | 0.78 | 1.12*** | 1.60*** | 0.99*** | 0.60 | 1.41** |
| | Germany | 0.42*** | 0.36*** | 0.43*** | 0.65*** | 0.46*** | 0.40*** | 0.37*** | 0.71*** | 0.39*** | 0.43*** | 0.63*** | −0.24** | 0.50*** | 0.25* | 0.10 | 0.65*** | −0.31 | 1.10*** |
| | Italy | 0.72*** | 1.21*** | 0.57*** | 0.62*** | 0.56*** | 0.48*** | 0.41*** | 0.70*** | 0.37*** | 0.53*** | 0.57*** | 0.17 | 0.94*** | 1.08*** | 0.99*** | 0.87*** | 0.55* | 1.73*** |
| | Japan | 1.00*** | 1.42*** | 0.89*** | 1.03*** | 0.87*** | 0.75*** | 0.78*** | 1.03*** | 0.81*** | 1.00*** | 0.93*** | 0.59*** | 1.14*** | 1.24*** | 1.40*** | 1.12*** | 0.77*** | 1.67*** |
| | DC | −0.03*** | −0.05*** | −0.02 | 0.01 | −0.00 | −0.01 | −0.01 | 0.00 | −0.02 | −0.01 | −0.01 | −0.03 | −0.07** | −0.09** | −0.01 | 0.03 | −0.02 | −0.09* |
| | PS | −0.13*** | −0.10 | −0.05 | −0.09 | −0.14* | −0.08 | −0.08 | −0.19* | −0.13 | −0.15* | −0.12 | −0.16 | −0.10 | −0.34 | −0.12 | −0.03 | −0.09 | −0.20 |
| Lexical complexity | Adj. $R^2$ | .33 | .33 | .20 | .12 | .18 | .17 | .15 | .18 | .17 | .34 | .12 | .18 | .13 | .19 | .52 | .24 | .23 | .44 |
| | Intercept | −3.44*** | −3.44*** | −2.74*** | −2.97*** | −2.81*** | −2.67*** | −2.67*** | −3.05*** | −2.76*** | −2.89*** | −2.83*** | −2.49*** | −2.96*** | −3.19*** | −3.74*** | −3.17*** | −2.62*** | −3.88*** |
| | China | 1.25*** | 1.25*** | 0.63*** | 0.82*** | 0.54*** | 0.52*** | 0.46** | 0.97*** | 0.65*** | 0.67*** | 0.73** | 0.51*** | 0.68 | 1.18*** | 1.68*** | 1.04*** | 0.52 | 0.96* |
| | Germany | 0.36*** | 0.36*** | 0.43*** | 0.65*** | 0.46*** | 0.41*** | 0.38*** | 0.71*** | 0.40*** | 0.42*** | 0.62*** | −0.23** | 0.50*** | 0.29** | 0.10 | 0.65*** | −0.30 | 1.10*** |
| | Italy | 0.87*** | 0.87*** | 0.48*** | 0.63*** | 0.55*** | 0.47*** | 0.41*** | 0.59*** | 0.29*** | 0.46*** | 0.46*** | 0.04 | 0.65*** | 0.48** | 0.83*** | 0.99*** | 0.54** | 1.01*** |
| | Japan | 1.32*** | 1.32*** | 0.88*** | 1.05*** | 0.89*** | 0.77*** | 0.81*** | 1.01*** | 0.80*** | 0.99*** | 0.90*** | 0.58*** | 1.10*** | 1.00*** | 1.34*** | 1.16*** | 0.78*** | 1.43*** |
| | LD | 0.00*** | 0.00*** | −0.00 | 0.00 | 0.00 | −0.00 | −0.00 | 0.00 | 0.00 | 0.00 | 0.00* | 0.00 | 0.00 | 0.00** | 0.00*** | 0.00 | −0.00 | 0.01** |

*Note.* Countries are coded as dummy variables, with the USA being used as the reference category. DC = Dale-Chall; PS = passive sentences %; LD = lexical diversity; Adj. $R^2$ = adjustive $R^2$.
*$p < .05$. **$p < .01$. ***$p < .00$.

containing more lexical diversity has a negligible positive effect on engagement. These outcomes are consistent with the notion of processing theory, in which easy-to-read messages tend to have higher engagement rates (Gkikas et al., 2022).

By advancing knowledge from psychological linguistics and with a particular focus on the language effects (Brouillet et al., 2023) of individual clusters in tourism, this research uncovered, on the one hand, that the combination of vibrant tourism pictures and simpler descriptions were more likely to be liked or commented on, and, on the other hand, that user engagement remained unaffected for posts linking cultural or historical spots with complex texts. The latter conclusion prompts further exploration into the necessity of a nuanced approach toward cultural or historical content. As these elements naturally evoke curiosity (Abraham, 2015), investigating how more detailed descriptions affect engagement by enhancing the understanding and appreciation of these rich cultural attributes becomes indispensable.

Also interesting is the fact that, in pictures featuring flora and fauna, atmospheric moods, and building landscapes, when complex texts were used, engagement rates decreased significantly. Although a recent study claimed that longer posts facilitate one's understanding on social media (Davis et al., 2019), it largely depends on the characteristics of the platform. In this case, since Instagram targets a primarily young audience (Newberry, 2023), the average user might prefer simple and direct messages instead. Specifically in destination marketing, while users might expect that cultural tourist spots could be described in a more complex manner for the reason of image-text congruence (Belanche et al., 2021; T. J. L.Van Rompay et al., 2010), there does not seem to be a preference over more sophisticated descriptions.

Furthermore, users disfavored non-cultural posts that contain intricate descriptions. Results therefore echo existing findings of short and narrative captions being effective in marketing destinations for younger users (Hooker & Cooper, 2022; Ismarizal & Kusumah, 2023) and of highly complex texts generating poor user engagement (Hine, 2015; Hong et al., 2023; McShane et al., 2019). To this end, Pancer and Poole (2016) found that sophisticated social media posts are also much less likely to be shared, and, according to Ko and Bowman (2020), posts that are deemed more complex receive fewer likes and comments. Intuitively, it might be assumed that algorithms filter out text-heavy content; however, in the case of picture-based posts on Instagram, the algorithm places more emphasis on users' interaction history and activities, prioritizing the perceived popularity of a post based on the frequency and speed of likes, comments, and shares (Mosseri, 2021). Consequently, concerning this research, reduced engagement has been attributed to memory-based processing difficulty, which resulted in the disruption of reading fluency (McShane et al., 2019). On social media platforms, where information overload and competition for attention is ubiquitous (Feng et al., 2015), users may easily disregard posts that require substantial processing effort.

Specific to the readability measures, the current results largely reverberated earlier findings. For instance, the Dale-Chall readability score was found to negatively correlate with the rate of post engagement (McShane et al., 2019). Passive voice density, similarly, affected reading negatively, whereas lexically diverse texts appeared to be favored by online users (Hine, 2015). Despite the highly comparable findings, care must be taken when it comes to lexical diversity as this study's data merely indicated a negligible positive effect on user engagement.

Taken together, this research demonstrates that processing fluency (e.g., Pancer et al., 2019) is a more important determinant than image-text congruence (e.g., T. J. L.Van Rompay et al., 2010) when aiming to market destinations on Instagram. Although user engagement was not hampered by complex descriptions in cultural or historical destination posts, no positive marketing effects could be noted either. Congruity between the intended image and description might have mitigated the negative effects of text complexity. On the contrary, simpler texts for vibrant tourism content attracted the most engagement as the posts were high in both processing fluency and image-text congruence. All in all, considering that destination posts target local citizens and international travelers alike, individuals or organizations working with social media should integrate simpler language to enhance readability and engagement to maximize advertising effects.

## Conclusion

### *Theoretical Contribution*

This research contributes to a deeper understanding of readability on social media by bridging the fields of data analytics, linguistics, and psychological esthetics with destination marketing. First, grounded in psychological esthetics, this research adopts processing fluency as its theoretical lens to offer empirical knowledge (Pancer et al., 2019) to the broader field of tourism marketing within the realm of social media. Furthermore, this research initiates a call to compare the effects between processing fluency and image-text congruity on user engagement. Past studies have implemented solely one of the two theories to explain behaviors of online users (Farace et al., 2020; McShane et al., 2019); however, the complementation of two psychological accounts should be able to paint a fuller picture from the perspectives of both destination marketers and tourists. This approach enables insights into the complex interactions between these stakeholders to be generated, shedding light on the intricacies of their relationship and ultimately providing more effective tourism marketing strategies.

On the whole, this study highlights intentionality as the key (Neuhofer et al., 2021) to delivering effective tourism marketing content—not just visually, but by integrating the art of textual elements based on different levels of information needed. In addition to the symphony of pictorial representations (Jia et al., 2021), leveraging user engagement requires the combination of structural and lexical elements in post captions. Thus, this research is novel in that it goes beyond the interplay between visual analysis and user engagement (Yu & Egger, 2021) by underlining the significance of textual characteristics on social media (Gkikas et al., 2022) in destination marketing (Li et al., 2023).

Moreover, distinct from relevant literature analyzing textual design artifacts by means of conventional measures (e.g., word counts and word frequency) (Gkikas et al., 2022), this study reinforces the potentiality of applying readability metrics from linguistics (Davis et al., 2019). Notably, despite the short and unstructured nature of social media posts (Egger & Yu, 2022) being unsuitable for commonly adopted scales such as the Flesch Kincaid metrics (Ismail et al., 2019), the findings showcase the usefulness of the Dale-Chall formula in addressing the readability of social media texts (Pancer et al., 2019). Through the examination of reading ease, text complexity, and user engagement, this research answers the call to evaluate processing fluency in the digital era (Davis et al., 2019; Gkikas et al., 2022; Pancer et al., 2019) and adds original knowledge to the understanding, research, and design of both pictures and texts with different levels of readability in the context of tourism (He et al., 2022; Fang et al., 2016; Yu & Egger, 2021).

### Practical Implications

In a nutshell, just as particular pictorial attributes (e.g., color) can moderate perceptions and intentions in tourism (Yu & Egger, 2021), this study seeks to unravel the specific linguistic elements that drive engagement. Additionally, it aims to dissect the complexities within captions and explore how readability influences varying levels of engagement rates based on different destination attributes derived from picture-based posts. Regarding the current dataset, it became evident that destination marketers seek to produce textual content that aligns with the intended product image on social media, such as applying sophisticated language to illustrate cultural richness. However, taking into account the advertising effectiveness reflected by user engagement, such efforts may not always pay off. It becomes even more challenging when complex language is applied to non-cultural or non-historical posts. Users may avoid spending time trying to comprehend and appreciate the beauty of the language and simply skip ahead to the next post (McShane et al., 2019), resulting in low engagement. Hence, understanding how readability influences engagement rates is pivotal since higher engagement enhances the likelihood of posts being amplified to a wider audience, thereby increasing exposure for DMOs. Given the high attentional competitiveness across social media (Feng et al., 2015), marketers are thus advised to simplify textual content regardless of the sophistication of the intended image of the tourist destination.

When exploring the current dataset further, it was also found that even slight changes in wording may have an immense impact on the likeliness of a post being liked and commented on, resulting in a "winner-takes-all" situation. For instance, in the category of "means of transport," @visittheusa published the following post captions alongside similarly amateur photos, but user engagement differed enormously:

1. *Continue your road trip on an estate tour in a hummer at Chawk Hill Winery in Sonoma County. This beautiful sprawling estate grows and bottles all of its own grapes*. (31 words, 6 likes, 0 comments)
2. *"Just your average street in Palm Springs! Borrowed bikes from our hotel and rode around for a morning. There's pretty much a view of the mountains at every turn." Sounds like a fantastic way to spend the morning!* (38 words, 2,138 likes, 28 comments)

According to the analyses, caption (1) was among the lowest in readability while caption (2) was one of the highest. With most observable attributes being similar (e.g., text length, narrative caption, professionalism of photography, etc.), and assuming equal advertising efforts from the same official Instagram account, it seems probable that textual complexity indeed plays a role in advertising effects. Although qualitative data analyses lie beyond the scope of the current study, this study offers compelling evidence pointing to the need for simpler post captions on social media. This significant outcome applies not only to official tourism organizations but to other tourism businesses that rely on effective marketing strategies as well.

### Limitations and Recommendations

This research is not without its limitations. Starting with the nature of social media posts, while acronyms and slang terms are often used, it remains ambiguous to what extent informal conversations influence structural complexity—especially considering that most of these words are not included on the list of frequently occurring terms. Likewise, though emojis may enhance the meaning of a sentence, they were purposely excluded in the analysis. Future studies are encouraged to explore the interaction between online terms/jargon, emojis, and text readability. At the same time, as the extracted posts were written by marketers with various linguistic/educational backgrounds, different grammatical mistakes, such as run-on sentences and comma splices, could be spotted. However, since reading on social media is completely different from "normal" reading (e.g., reading textbooks), researchers are recommended to explore whether certain errors may have a negative impact on not just readability itself but also other potential aspects, for instance, destination image or credibility. In this study, grammar issues such as run-on sentences are assumed to complicate readability and should therefore be accounted for.

In spite of multiple metrics having been used in this study to cater for both syntactic and lexical complexity, cross-disciplinary collaboration with linguists is highly suggested to develop readability measurements that fit social media content the best. Additionally, although image clustering was conducted as the first step to facilitating a more coherent examination of the content, definitively attributing user engagement solely to either the photo or the caption remains challenging. Future research may explore more sophisticated methodologies to disentangling the individual contributions of visuals and texts in driving user engagement on social media platforms. Furthermore, as one of the first studies providing additional insights on the topic of readability and social media posts, future investigations can consider taking cultural differences into account. Finally, this research relied on Instagram for its dataset; for this reason, scholars should also extract and compare posts shared on other platforms in the context of tourism.

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